

RAUMA KYLMAPIHLAJA, Finland

WMO: 027610

Lat: 61.1447N

Lon: 21.3028E

Elev: 95

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.5	7.2	-2.7	4.8	2.1	3.4	6.6	8.2	49.7	31.8	45.3	32.6	15.2	70	0.923

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	74.5	67.9	71.3	65.6	68.9	63.9	69.1	73.4	66.6	70.1	64.6	68.2	14.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	100.8	72.1	65.2	93.6	69.6	62.9	86.2	67.2	33.2	73.9	31.3	69.8	29.8	68.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.8	35.9	32.5	DB	2.6	78.6	10.0	5.4	-4.6	82.5	-10.5	85.7	-16.1	88.7	-23.3	92.6	(4)
(5)			WB	1.0	69.4	9.6	3.2	-5.9	71.7	-11.5	73.6	-16.9	75.5	-23.9	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	25.9	26.7	30.6	37.7	48.2	55.5	62.4	62.2	55.5	45.2	37.9	32.0		
	DBStd	14.31	8.39	9.09	6.00	4.43	6.02	5.29	5.30	4.38	4.58	5.20	6.63	8.52		
	HDD50	3577	748	653	602	370	105	9	0	0	11	160	362	557		
	HDD65	7938	1213	1073	1067	818	520	290	116	107	286	614	812	1022		
	CDD50	1176	0	0	0	2	50	174	384	380	175	11	0	0		
	CDD65	62	0	0	0	0	0	5	35	21	1	0	0	0		
	CDH74	120	0	0	0	0	1	10	56	53	0	0	0	0		
	CDH80	12	0	0	0	0	0	0	4	8	0	0	0	0		
	WSAvg	17.6	18.9	18.3	18.4	16.1	15.9	15.7	15.1	16.1	17.2	19.4	19.1	20.8		
	PrecAvg	24.8	2.0	1.5	1.3	1.3	1.5	2.2	2.7	2.8	2.3	2.7	2.4	2.3		
	PrecMax	32.2	3.9	3.1	2.3	2.4	3.3	4.3	5.3	5.4	5.0	6.3	5.5	4.3		
	PrecMin	20.2	0.3	0.1	0.2	0.0	0.2	0.7	0.1	1.0	0.5	0.3	0.4	0.3		
	PrecStd	3.0	0.9	0.8	0.5	0.6	0.8	0.9	1.3	1.3	1.4	1.4	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	38.4	44.3	56.8	69.5	75.5	79.1	80.2	69.0	57.1	50.2	45.4		
		MCWB	36.9	36.5	37.8	49.0	58.8	63.0	70.3	69.9	62.2	54.0	48.6	43.9		
	2%	DB	37.6	37.3	40.1	49.4	63.3	69.3	75.0	73.9	64.7	54.9	48.8	43.3		
		MCWB	36.2	35.7	37.4	44.4	56.2	62.2	69.0	67.7	60.5	52.3	47.7	41.7		
	5%	DB	36.3	36.3	38.2	45.8	59.5	65.9	72.2	70.6	62.8	53.1	47.1	41.6		
		MCWB	35.1	35.1	36.2	42.1	54.4	60.8	67.5	64.7	59.2	50.9	45.9	39.9		
	10%	DB	35.2	35.4	36.9	43.5	56.1	62.9	69.8	67.9	61.2	51.7	45.5	40.3		
		MCWB	34.1	34.4	35.3	40.4	51.9	58.2	65.7	62.9	58.1	49.6	44.1	38.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	37.3	39.4	48.9	60.3	66.2	71.7	71.5	62.9	54.6	49.1	44.4		
		MCDB	38.2	37.7	42.0	55.8	67.0	70.6	76.6	77.4	67.1	56.0	49.7	45.1		
	2%	WB	36.5	36.1	37.8	44.9	57.6	63.0	70.0	68.1	61.1	53.1	48.0	41.9		
		MCDB	37.1	36.7	39.4	48.6	62.4	67.7	74.1	72.9	64.1	54.7	48.8	42.9		
	5%	WB	35.3	35.3	36.6	42.8	55.0	61.0	68.1	65.6	59.8	51.5	46.2	40.2		
		MCDB	36.1	36.0	37.9	45.3	58.9	65.7	71.7	69.6	62.4	52.7	47.0	41.3		
	10%	WB	34.2	34.5	35.6	40.9	52.3	58.9	66.1	63.9	58.7	49.9	44.3	38.9		
		MCDB	35.0	35.3	36.7	43.0	55.9	62.5	69.0	67.2	60.9	51.2	45.3	40.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	6.3	7.1	7.0	8.0	7.4	7.3	6.8	6.1	5.0	4.7	5.2		
		MCDBR	4.5	4.2	6.3	10.7	13.6	12.3	10.8	10.2	8.0	5.5	4.5	4.7		
		MCWBR	4.3	3.7	4.8	7.0	7.6	7.0	6.2	6.1	5.3	4.9	4.3	5.0		
	5% WB	MCDBR	4.4	4.3	5.8	10.2	13.5	11.3	9.5	8.5	7.4	5.3	4.8	4.7		
		MCWBR	4.3	3.9	4.8	7.0	8.1	6.9	6.0	5.9	5.2	5.2	4.7	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.310	0.327	0.367	0.365	0.356	0.383	0.370	0.345	0.328	0.288	0.361		
	taud		2.152	2.249	2.320	2.327	2.417	2.478	2.427	2.451	2.503	2.401	2.196	2.560		
	Ebn at Noon		144	210	251	262	272	277	265	259	246	204	133	95		
	Edh at Noon		12	21	27	33	32	31	32	29	23	18	11	8		
(44) All-Sky Solar Radiation	RadAvg		64	247	736	1256	1756	1933	1781	1386	827	344	89	38		
	RadStd		9	47	83	122	144	118	157	121	87	55	16	4		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	-212	N/A	N/A	N/A

Nomenclature: See separate page